Design Document: Mach-Zehnder Interferometer

JIE SUN

Abstract: The primary objective of this design report is to fabricate and test an unbalanced Mach-Zehnder Interferometer (MZI) circuit to experimentally extract the waveguide group index (n_g) near the 1550 nm wavelength. By testing six MZI variations with incrementally increasing imbalance lengths (ΔL), we will observe the trend of the Free Spectral Range (FSR) varying with ΔL and compare the experimentally extracted n_g against Lumerical FDTD and MODE simulations.

1. Introduction

Silicon photonics provides a scalable platform for dense photonic integrated circuits across optical communications, sensing, and computing applications. The fundamental routing backbone of these circuits relies on high-index-contrast silicon waveguides to efficiently confine and propagate light, while fiber grating couplers provide the essential vertical optical interface to couple light between off-chip single-mode fibers and sub-micron on-chip devices.

The primary design objective of this project is to construct an unbalanced Mach-Zehnder Interferometer (MZI) circuit using standard strip waveguides and TE grating couplers to experimentally extract the waveguide group index (n_g) near the 1550 nm operating wavelength. By varying the physical path length difference (ΔL) across six distinct designs, we will systematically measure the relationship between the Free Spectral Range (FSR) and ΔL, validate our Lumerical simulation models, and establish a baseline for manufacturing variability.

2. Theory

2.1 Interferometer Theory and Transfer Function

The unbalanced MZI splits an incoming optical wave of power I_{in} equally into two arms using a 3dB 50/50 splitter (ideal Y-branch). The physical path length difference between the two arms is $\Delta L = L1 - L2$. Recombining the signals at an identical Y-branch produces interference governed by the phase difference $\Delta\phi(\lambda)$:

$$\Delta\phi(\lambda) = 2\pi \frac{n_{eff}(\lambda)}{\lambda}\Delta L$$

Assuming lossless propagation and ideal 50/50 Y-branches, the optical power transfer function $T(\lambda) = \frac{I_{out}(\lambda)}{I_{in}(\lambda)}$ is given by:

$$T(\lambda) = cos^2(\frac{\Delta\phi(\lambda)}{2}) = \frac{1}{2}[1 + cos(2\pi \frac{n_{eff}(\lambda)\Delta L}{\lambda})]$$

When waveguide propagation loss α (in m^{-1}) is included, the power transfer function becomes:

$$T(\lambda) = \frac{1}{4}e^{-\alpha L_1}[1 + e^{-\alpha\Delta L} + 2e^{-\frac{1}{2}\alpha\Delta L}cos(2\pi \frac{n_{eff}(\lambda)\Delta L}{\lambda})]$$

2.2 Derivation of Group Index Equation from Free Spectral Range

The Free Spectral Range (FSR) is defined as the spacing between consecutive transmission maxima (or minima) in the optical spectrum. Let λ_1 and λ_2 be two adjacent peak wavelengths, where $\lambda_2 > \lambda_1$ and $FSR = (\lambda_2 - \lambda_1) = \Delta\lambda$.

The phase condition for consecutive transmission peaks corresponding to integer interference orders m and $m - 1$ is:

$$\Delta\phi(\lambda_1) = \frac{2\pi n_{eff}(\lambda_1)}{\lambda_1}\Delta L = 2m\pi$$

$$\Delta\phi(\lambda_2) = \frac{2\pi n_{eff}(\lambda_2)}{\lambda_2}\Delta L = 2(m-1)\pi$$

Subtracting the two phase expressions yields:

$$\Delta\phi(\lambda_1) - \Delta\phi(\lambda_2) = 2\pi$$

To express this phase change in terms of dispersion, we apply a first-order Taylor series expansion of $\Delta\phi(\lambda)$ around λ_1:

$$\Delta\phi(\lambda_2) \approx \Delta\phi(\lambda_1) + \left.\frac{d\Delta\phi}{d\lambda}\right|_{\lambda_1}(\lambda_2 - \lambda_1)$$

Evaluating the derivative of phase with respect to wavelength:

$$\frac{d\Delta\phi}{d\lambda} = \frac{d}{d\lambda}\left(\frac{2\pi n_{eff}(\lambda)\Delta L}{\lambda}\right) = 2\pi\Delta L\left(\frac{\frac{dn_{eff}}{d\lambda}\cdot\lambda - n_{eff}(\lambda)}{\lambda^2}\right) = -\frac{2\pi\Delta L}{\lambda^2}\left(n_{eff}(\lambda) - \lambda\frac{dn_{eff}}{d\lambda}\right)$$

Recalling the definition of waveguide group index $n_g(\lambda) = n_{eff}(\lambda) - \lambda\frac{dn_{eff}}{d\lambda}$:

$$\frac{d\Delta\phi}{d\lambda} = -\frac{2\pi\Delta L\cdot n_g(\lambda)}{\lambda^2}$$

Substituting this derivative into the Taylor expansion and setting $|\Delta\phi(\lambda_2) - \Delta\phi(\lambda_1)| = 2\pi$:

$$\frac{2\pi\Delta L\cdot n_g}{\lambda_0^2}\Delta\lambda = 2\pi$$

Solving for the group index n_g:

$$n_g = \frac{\lambda_1\lambda_2}{\Delta L\cdot \mathrm{FSR}} \approx \frac{\lambda_0^2}{\Delta L\cdot \mathrm{FSR}}$$

Where $\lambda_0 = \sqrt{\lambda_1\lambda_2} \approx \frac{\lambda_1+\lambda_2}{2}$ is the center wavelength between adjacent peaks.

3. Model and Simulation

3.1 Waveguide Geometry and Mode Profile

3.1.1 Waveguide Parameters

- **Core Material**: Silicon (Si, $n \approx 3.47$ at 1550 nm)
- **Cladding Material**: Silicon Dioxide (SiO_2, $n \approx 1.448$ at 1550 nm)

- **Waveguide Cross-Section**: Width $w = 500nm$, Height $h = 220nm$
- **Polarization**: Fundamental TE (TE_{00})

3.1.2 Mode Profile Simulation

The modal distribution of the fundamental TE mode was simulated in Lumerical MODE Solutions using the Finite Difference Eigenmode (FDE) solver. The fundamental TE mode exhibits strong dominant transverse electric field polarization E_x, with the optical energy tightly confined within the 500 $nm \times 220$ nm silicon core.

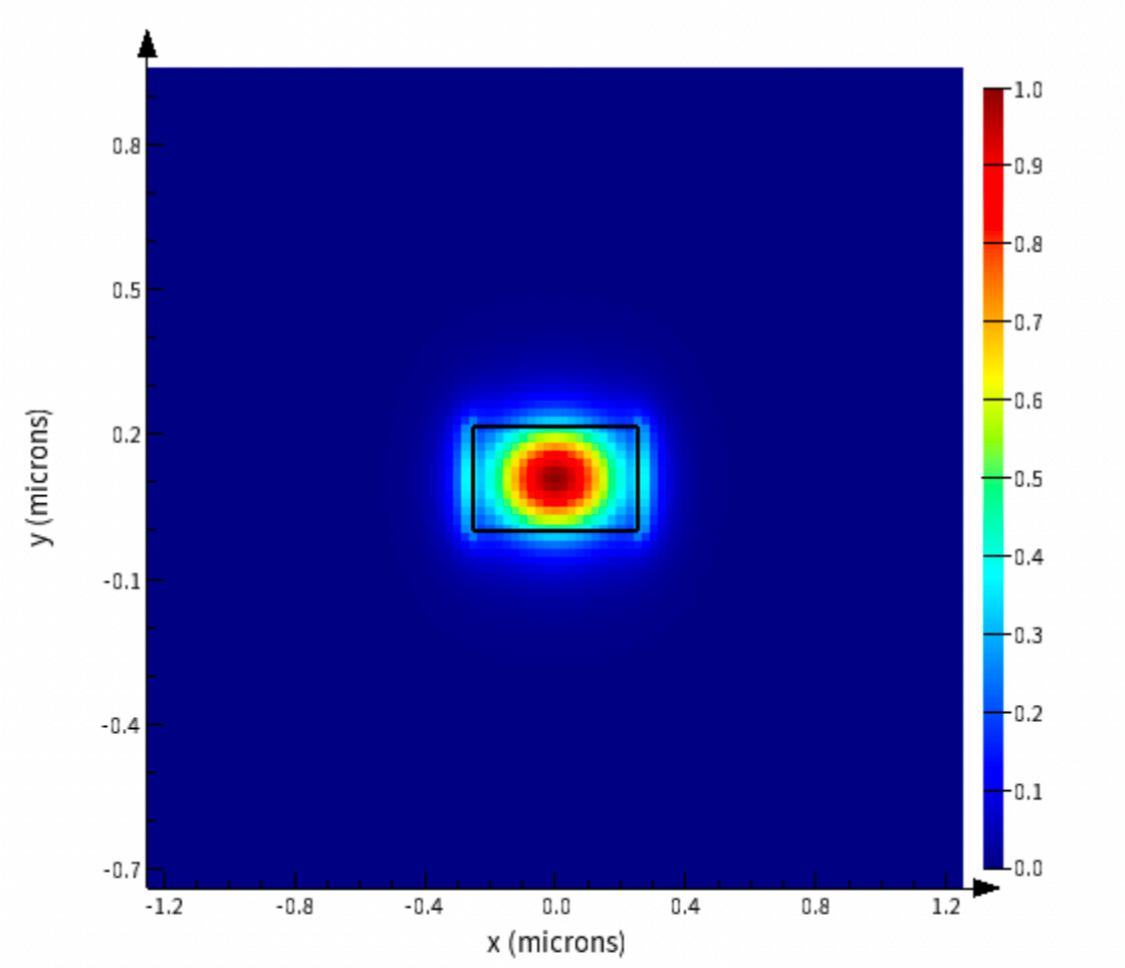


Figure 1: Lumerical Mode FDE Cross-Sectional Field Profile for a 500 nm x220 nm silicon strip waveguide at $\lambda = 1550$ nm

3.2 Compact Model and Waveguide Dispersion

3.2.1 Effective Index (n_{eff}) and Group Index (n_g) Curves

Using the frequency sweep feature in Lumerical MODE over 1.5 μm to 1.6 μm wavelength range, the wavelength dependence of the effective index (n_{eff}) and group index (n_g) was calculated.

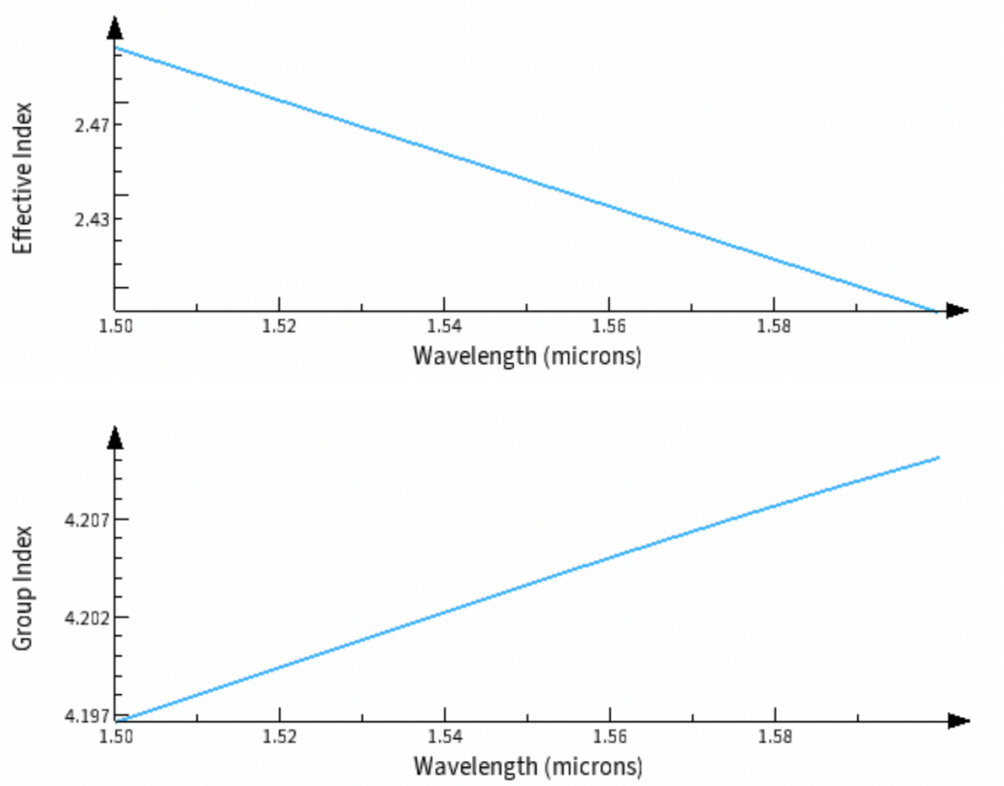


Figure 2: Simulated Effective Index (n_{eff}) and Group Index (n_g) versus wavelength ($\lambda = 1.5\ \mu m\ to\ 1.6\ \mu m$) for Fundamental TE mode

At the target wavelength $\lambda_0 = 1550\ nm$, the nominal values determined from simulation are:

•Effective Index (n_{eff}): $n_{eff} \approx 2.44$

•Group Index (n_g): $n_g \approx 4.20$

3.2.2 Compact Model Polynomial Expression

To enable fast circuit-level simulations in Lumerical INTERCONNECT, $n_{eff}(\lambda)$ is expanded as a second-order polynomial centered at $\lambda_0 = 1.55\ \mu$m:

$$n_{eff}(\lambda) = n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2$$

Where λ is specified in micrometers (μm), and the polynomial coefficients obtained from the Lumerical polynomial fit script are:

$$n_0 = n_{eff}(\lambda_0) \approx 2.4412$$

$$n_1 = \left.\frac{d n_{eff}}{d\lambda}\right|_{\lambda_0} \approx -1.1341\ \mu\text{m}^{-1}$$

$$n_2 = \left.\frac{1}{2}\frac{d^2 n_{eff}}{d\lambda^2}\right|_{\lambda_0} \approx 0.1250\ \mu\text{m}^{-2}$$

The group index $n_g(\lambda)$ is related to the effective index by:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda\frac{d n_{eff}(\lambda)}{d\lambda}$$

3.3. Extract Group Index

3.3.1 Group Index Through Simulation.

Using the baseline simulation parameters ($\Delta L = 40\ \mu$m, $\lambda_0 = 1.55\ \mu$m, simulated $n_g = 4.20$):

$$\text{FSR}_{sim} = \frac{(1.55\times10^{-6})^2}{4.20\times(40\times10^{-6})} = 1.4315\times10^{-8}\ \text{m} = 14.315\ \text{nm}$$

Re-extracting n_g from FSR_{sim} :

$$n_g = \frac{(1.55\times10^{-6})^2}{(40\times10^{-6})\times(14.315\times10^{-9})} = 4.20$$

This confirms the mathematical consistency of the extraction formula.

3.3.2 Experimental Extraction Methodology

- Peak Detection: Measure the transmission spectrum $T(\lambda)$ across 1500 nm − 1600 nm. Find all peak wavelengths λ_m
- FSR Calculation: Compute local peak-to-peak spacing $\text{FSR}_m = \lambda_{m+1} - \lambda_m$
- Linear Regression: Plot interference peak order m versus optical frequency $\nu_m = c/\lambda_m$. The slope of $m(\nu)$ equals $\frac{n_g\Delta L}{c}$, allowing n_g to be extracted via linear regression over the full spectral band to minimize experimental noise.

3.4 Parameter Variations and Expected Performance

To isolate ΔL as the single variable, six MZI layout configurations are designed using uniform 500 nm × 220 nm TE strip waveguides, Y-branch splitters, and bend radii at 5 μm to ensure negligible bend loss ($< 0.1\,\%$).

All target FSR values strictly satisfy the condition of remaining within the ~ 50 nm measurement bandwidth of the optical test system.

Design	Imbalance Length (ΔL)	Waveguide Width	Waveguide Height	Polarization	Target Wavelength(λ_0)	Expected FSR (nm)
MZI 1	25 μm	500 nm	220 nm	TE	1550 nm	22.88 nm
MZI 2	50 μm	500 nm	220 nm	TE	1550 nm	11.44 nm
MZI 3	75 μm	500 nm	220 nm	TE	1550 nm	7.63 nm
MZI 4	100 μm	500 nm	220 nm	TE	1550 nm	5.72 nm
MZI 5	125 μm	500 nm	220 nm	TE	1550 nm	4.58 nm
MZI 6	150 μm	500 nm	220 nm	TE	1550 nm	3.81 nm

Table 1. Six MZI layout configurations

3.5 Transmission Spectrum Simulations

Using the compact waveguide model inside Lumerical INTERCONNECT, circuit-level simulations were executed for the unbalanced MZI structures.

The simulated transmission exhibits sinusoidal interference fringes superimposed on the Gaussian bandpass response of the input/output fiber grating couplers. As ΔL increases from 25 μm to 150 μm, the fringe density increases, corresponding to the expected reduction in FSR from 22.88 nm to 3.81 nm.

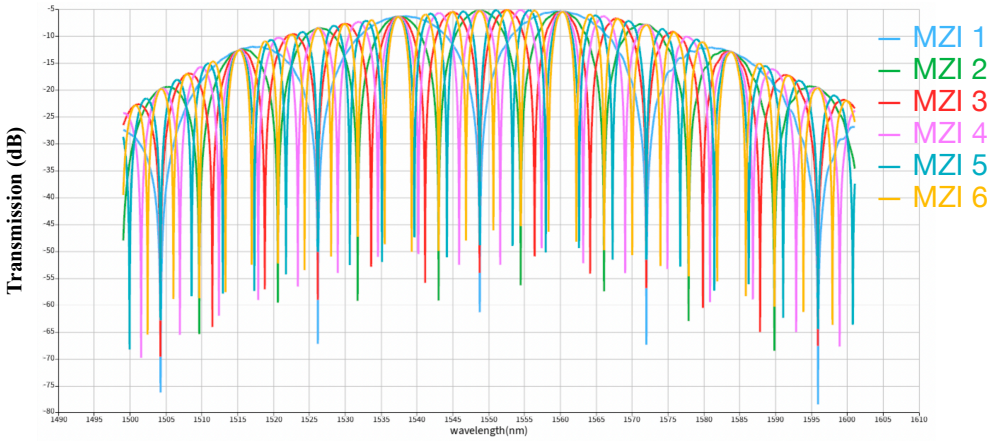


Figure 3: Simulated Transmission Spectrum (dB) vs. Wavelength for MZI

4. Fabrication

4.1 Manufacturing Variability and Corner Analysis Methodology

In silicon photonics fabrication, standard electron-beam lithography and etching processes introduce dimensional variations that deviate from the ideal layout. For silicon-on-insulator (SOI) strip waveguides, the dominant manufacturing variations are fluctuations in the

waveguide core width (Δw) due to lithography and etch biases, and variations in the top silicon device layer thickness (Δh) inherent to the SOI wafer manufacturing process.

To quantify the impact of these manufacturing tolerances on our unbalanced Mach-Zehnder Interferometer (MZI) circuits, a corner analysis is performed. The nominal waveguide dimensions are defined as $w = 500$ nm and $h = 220$ nm. Assume that the waveguide thickness varies between 215.3 nm and 223.1 nm while width varies between 470 nm and 510 nm, five eigensolver simulations are executed in Lumerical MODE to capture the extreme "corners" of the fabrication window alongside the nominal design:

- Nominal: $w = 500$ nm, $h = 220$ nm
- Corner 1: $w = 470$ nm, $h = 215.3$ nm
- Corner 2: $w = 510$ nm, $h = 215.3$ nm
- Corner 3: $w = 470$ nm, $h = 223.1$ nm
- Corner 4: $w = 510$ nm, $h = 223.1$ nm

4.2 Impact on Waveguide Parameters (n_{eff} and n_g)

Because the fundamental TE mode is highly confined within the $500\text{ nm} \times 220\text{ nm}$ silicon core, any geometric variation alters the mode confinement, subsequently shifting the effective index (n_{eff}) and group index (n_g). A reduction in the waveguide cross-section pushes a larger fraction of the optical mode field into the oxide cladding, lowering n_{eff} and altering structural dispersion (which directly impacts n_g).

Using Lumerical MODE, compact models for the waveguides at each corner are generated to extract these parameters at the target wavelength of $\lambda_0 = 1550$ nm. The anticipated variations are recorded in Table 2.

Design Corner	Width (w)	Height (h)	Effective Index (n_{eff})	Group Index (n_g)
Nominal	500 nm	220 nm	2.44682	4.20367
Corner1	470 nm	215.3 nm	2.37202	4.25227
Corner2	510 nm	215.3 nm	2.4337	4.17875
Corner3	470 nm	223.1 nm	2.40615	4.26067
Corner4	510 nm	223.1 nm	2.47375	4.18781

Table 2. Simulated Waveguide Parameter Variations at $\lambda_0 = 1550$ nm

4.3 Impact on FSR and Central Wavelength

The dimensional fluctuations evaluated in the corner analysis propagate directly to the macroscopic performance of the six MZI configurations. We evaluate this impact across two primary performance metrics:

4.3.1 Free Spectral Range (FSR) Variability:

The FSR is inversely proportional to the group index, defined by $FSR \approx \frac{\lambda_0^2}{\Delta L \cdot n_g}$. Because n_g fluctuates with waveguide geometry, the measured FSR will deviate from the nominal targets. The expected FSR boundaries for MZIs 1 through 6, calculated using the extreme n_g values extracted from the Lumerical corner analysis, are summarized in Table 3.

Design	Imbalance Length(ΔL)	Nominal Expected FSR	FSR Minimum (Using Max n_g)	FSR Maximum (Using Min n_g)
MZI 1	25 μm	22.88 nm	22.56 nm	23.00 nm
MZI 2	50 μm	11.44 nm	11.28 nm	11.50 nm
MZI 3	75 μm	7.63 nm	7.52 nm	7.67 nm
MZI 4	100 μm	5.72 nm	5.64 nm	5.75 nm
MZI 5	125 μm	4.58 nm	4.51 nm	4.60 nm
MZI 6	150 μm	3.81 nm	3.76 nm	3.83 nm

Table 3. MZI Free Spectral Range (FSR) Corner Analysis Estimates

4.3.2 Central Wavelength Shift:

While variations in n_g dictate the spacing between interference fringes, variations in the effective index (n_{eff}) dictate the absolute spectral position of the transmission peaks. The phase difference governing the interference condition is highly sensitive to n_{eff}:

$$\Delta\phi(\lambda) = 2\pi \frac{n_{eff}(\lambda)}{\lambda} \Delta L$$

Even a nanometer-scale deviation in the waveguide cross-section alters n_{eff}, resulting in a substantial accumulated phase error over the imbalance length ΔL. This causes the entire transmission spectrum to shift laterally across the wavelength band.

To quantify this, circuit-level simulations were executed in Lumerical INTERCONNECT for MZI 1 ($\Delta L = 25\ \mu m$) using the compact waveguide models from nominal and the four corner conditions. As shown in Figure 4, the transmission spectra exhibit significant spectral drifting.

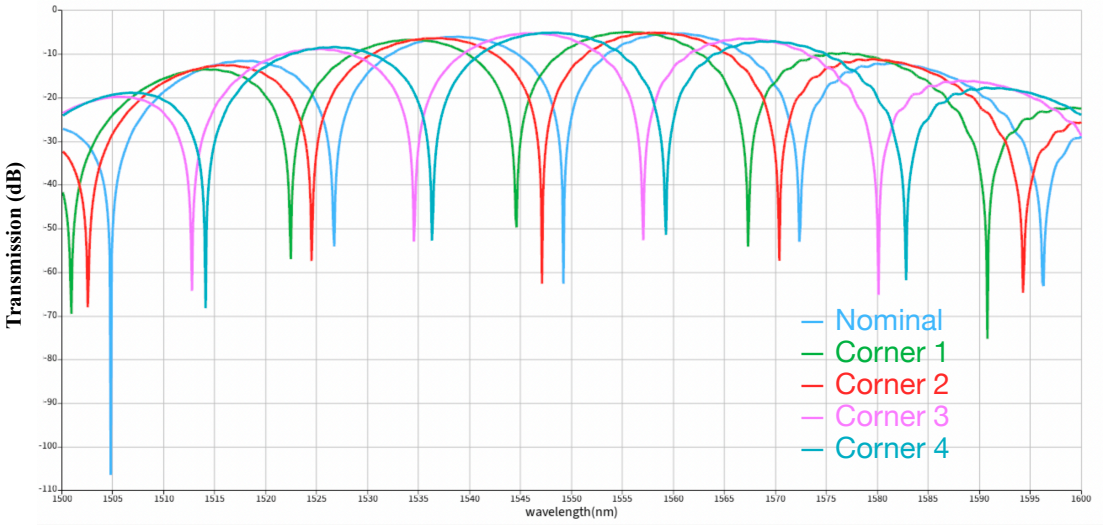


Figure 4: Simulated transmission spectrum of MZI 1 ($\Delta L = 25\ \mu m$) across five fabrication corners, illustrating central wavelength drift near 1550 nm due to geometric variations.

By isolating a specific transmission peak near the nominal 1550 nm target, we can extract the exact wavelength shift ($\Delta\lambda$) caused by the dimensional tolerances. The resulting peak variations are summarized in Table 4.

Design Corner	Width (w)	Height (h)	Wavelength near 1550 nm (λ_{peak})	Wavelength Shift ($\Delta\lambda$ from Nominal)
Nominal	500 nm	220 nm	1549.19 nm	0 nm
Corner1	470 nm	215.3 nm	1544.6 nm	-4.59 nm
Corner2	510 nm	215.3 nm	1547.09 nm	-2.1 nm
Corner3	470 nm	223.1 nm	1557.03 nm	7.84 nm
Corner4	510 nm	223.1 nm	1559.25 nm	10.06 nm

Table 4. Central Wavelength Shift for MZI 1 ($\Delta L = 25\ \mu m$) Across Fabrication Corners

This analysis demonstrates that unbalanced Mach-Zehnder Interferometers are highly sensitive to manufacturing tolerances. While the FSR remains relatively stable, the absolute peak positions are difficult to predict precisely without active thermal tuning or post-fabrication trimming.

5. Experiment data

(To be completed)

6. Analysis

(To be completed)

7. Conclusion

(To be completed)

8. References

1. Lukas Chrostowski, Michael Hochberg, Silicon Photonics Design: From Devices to Systems, Cambridge University Press, 2015
2. Lukas Chrostowski, Silicon Photonics Design, Fabrication and Data Analysis EDX Course Material, 2026